

Autonomic Pharmacology

Cholinergic and Adrenergic
Drugs

Pharmacology — Lecture 7
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Outline

- Neurotransmission in the ANS
- Receptors
- Cholinergic drugs
- Adrenergic drugs
- Case discussion

Neurotransmitters of the autonomic nervous system

Acetylcholine is the neurotransmitter of all preganglionic autonomic fibers and of postganglionic parasympathetic fibers. Norepinephrine is the neurotransmitter of most postganglionic sympathetic fibers. Sweat glands are innervated by sympathetic cholinergic fibers.

Acetylcholine is synthesized from choline and acetyl-CoA by choline acetyltransferase. It is hydrolyzed in the synaptic cleft by acetylcholinesterase.

Autonomic receptors

Receptor	G protein	Second messenger	Major location
M1	Gq	Increased IP3 and DAG	CNS and gastric parietal cells
M2	Gi	Decreased cAMP	Heart
M3	Gq	Increased IP3 and DAG	Smooth muscle and exocrine glands
Alpha-1	Gq	Increased IP3 and DAG	Vascular smooth muscle
Alpha-2	Gi	Decreased cAMP	Presynaptic nerve terminals
Beta-1	Gs	Increased cAMP	Heart and juxtaglomerular cells
Beta-2	Gs	Increased cAMP	Bronchial smooth muscle

Direct-acting cholinergic agonists

Bethanechol

Bethanechol is a muscarinic agonist that is resistant to acetylcholinesterase. It is used to treat postoperative urinary retention.

Pilocarpine

Pilocarpine is a muscarinic agonist used to treat glaucoma. It causes miosis and contraction of the ciliary muscle.

Indirect-acting agonists (cholinesterase inhibitors)

- Neostigmine is a reversible cholinesterase inhibitor that does not cross the blood–brain barrier.
 - It is used to reverse neuromuscular blockade and to treat myasthenia gravis.
- Physostigmine is a tertiary amine that crosses the blood–brain barrier. It is used to treat anticholinergic toxicity.
- Organophosphates irreversibly inhibit acetylcholinesterase.
- Pralidoxime regenerates acetylcholinesterase if given before aging of the enzyme.

Organophosphate poisoning

Muscarinic signs of organophosphate poisoning

Diarrhea

Urination

Miosis

Bronchospasm

Bradycardia

Lacrimation

Salivation

Atropine is given to block the muscarinic effects of organophosphate poisoning.

Pralidoxime is given to reverse the nicotinic effects, such as muscle weakness.

Case discussion

A 45-year-old farmer presents with pinpoint pupils, excessive salivation and bradycardia.

What is the most likely cause?

Which antidote would you give first?

Muscarinic antagonists

Atropine

Atropine is a competitive antagonist at muscarinic receptors. It increases heart rate and causes mydriasis. It is used to treat symptomatic bradycardia.

Ipratropium and scopolamine

Ipratropium is an inhaled muscarinic antagonist used in chronic obstructive pulmonary disease. Scopolamine is used to prevent motion sickness.

Adverse effects of muscarinic antagonists include dry mouth, blurred vision, urinary retention, constipation and hyperthermia.

Adrenergic agonists

Drug	Receptor selectivity	Clinical use
Epinephrine	Alpha and beta	Anaphylaxis
Norepinephrine	Alpha-1, alpha-2 and beta-1	Septic shock
Dobutamine	Beta-1 more than beta-2	Acute heart failure
Phenylephrine	Alpha-1	Nasal decongestion
Albuterol	Beta-2	Acute asthma
Clonidine	Alpha-2	Hypertension

Epinephrine and clonidine

Epinephrine

Epinephrine is the drug of choice for anaphylaxis. At low doses it has predominantly beta effects. At high doses alpha-1 effects predominate and cause vasoconstriction.

Clonidine

Clonidine is a centrally acting alpha-2 agonist. It decreases sympathetic outflow from the brainstem. Abrupt withdrawal of clonidine may cause rebound hypertension.

Indirect sympathomimetics

- Amphetamine releases stored catecholamines from nerve terminals.
- Cocaine blocks the reuptake of norepinephrine, dopamine and serotonin.
- Cocaine is the only local anesthetic that causes vasoconstriction.

Alpha blockers

- Prazosin is a selective alpha-1 antagonist used to treat hypertension and benign prostatic hyperplasia. Its main adverse effect is first-dose orthostatic hypotension.
- Phenoxybenzamine is an irreversible nonselective alpha antagonist used to treat pheochromocytoma before surgery.

Beta blockers

Drug	Selectivity	Additional use
Propranolol	Nonselective (beta-1 and beta-2)	Migraine prophylaxis
Metoprolol	Beta-1 selective	Heart failure
Timolol	Nonselective	Glaucoma

Beta blockers decrease heart rate and myocardial contractility.

Nonselective beta blockers can precipitate bronchospasm in patients with asthma.

Beta blockers may mask the symptoms of hypoglycemia in diabetic patients.

Summary

- Cholinergic agonists → parasympathetic effects
- Antagonists → the opposite
- Know the receptor table!

References

- Katzung BG. Basic and Clinical Pharmacology. 15th ed. McGraw-Hill; 2021.
- Whalen K. Lippincott Illustrated Reviews: Pharmacology. 7th ed.